Factor-base designs: another prime, structured targets and m = 5 (run-20261020T1555Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only: nothing here prices a pipeline, reports S or shows a speedup. **Protocol:** PREREGISTRATION.md Amendment 9 (registered in commit `ad791be5b` before any P4, Z1 or M5 row) and Amendment 10 (registered in commit `e04733876` before the Z1 rows it adds), with Amendments 1–8 for everything they do not change.

What this run takes from "not yet shown"

The Amendment 8 report listed four kinds of instance the study had not touched. This run measures three of them: **P4**, a prime of another 2-adic shape; **Z1**, structured targets; **M5**, m = 5. The fourth, other curve models (Edwards, binary-field curves), is left out, for the reasons registered in Amendment 9: each needs its own summation polynomial or field and solver, so it would not be measured by this study's instrument.

Run

- **Commands:** P4: `fb_designs_slope run --prime p4 --arms A0,A11 --m 2 --t-max 8 / --m 3 --t-max 4 --budget 300 --vmem-kb 13000000`. Z1: the same with `--target-structure xr0` on p_1, `--m 2 --t-max 9` and `--m 3 --t-max 4`. M5: `run --arms A0 --m 5 --t-min 3 --t-max 3 --budget 300`. The sequence is `runs/amendment9-sequence.sh`.
- **Commits:** P4, Z1 (m = 3) and M5 at `ad791be5b0f7bda1f000814656117f277a0bf73f` Z1 (m = 2, t ≥ 3, Amendment 10) at `e047338764dace77e3d91fa5397349ef6e37dd8d`, recorded in each row.
- **Host:** "vm", Intel Xeon @ 2.80 GHz, 4 cores, 15 GB, no swap, a Claude Code cloud container. One system at a time. Same limits as the first run: 300 s budget, 420 s hard kill, `ulimit -v 13000000`, 2 planted targets per cell.
- **Window (UTC, 2026-10-20):** P4 15:55–15:58Z; Z1 15:58–15:59Z and, after Amendment 10, 16:10–16:18Z; M5 15:59–16:10Z.
- **p_4:** 10 007 297 = $2^8 \cdot 39\,091 + 1$, with $p_4 - 1 = 2^8 \cdot 13 \cdot 31 \cdot 97$: 2-adic valuation 8 ($p_1$, p_2, p_3 have 18, 27, 13), so t ≤ 8, and a smooth odd part.
- **Records:** `runs/P4-*`, `runs/Z1-*`, `runs/M5-*` and `runs/analysis-a9.txt` (native `analyze --a9`).

Graph

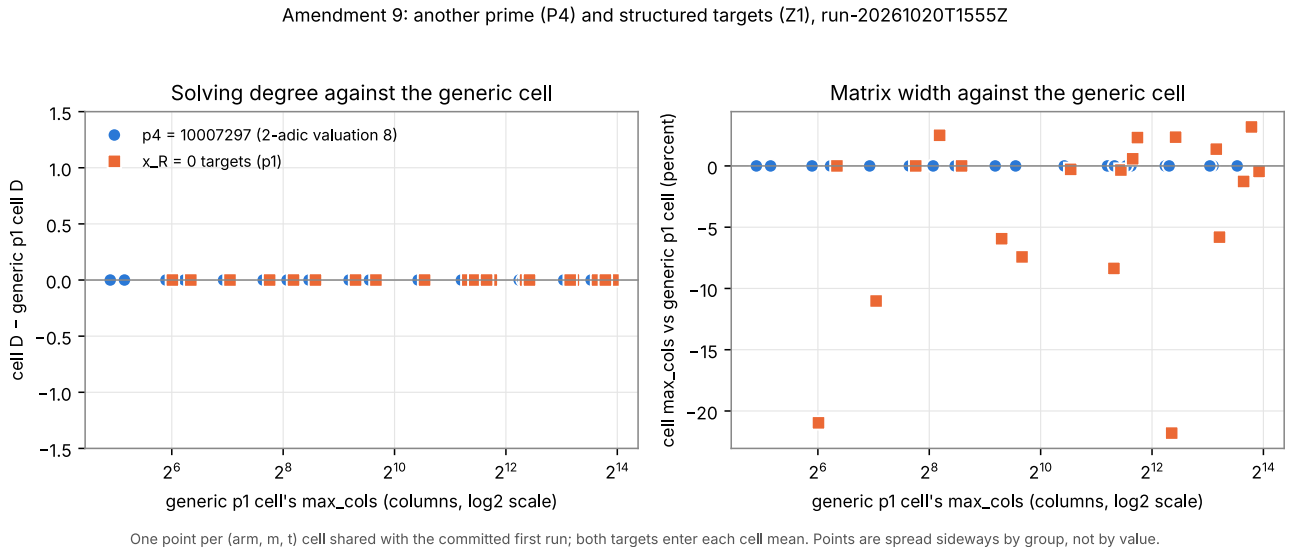


Figure 1: Per (arm, m, t) cell shared with the committed first run (generic p_1): the cell's D minus the generic D (left) and its `max_cols` against the generic cell's, in percent (right), for P4 (p_4, generic targets) and Z1 (p_1, $x_R = 0$ targets). 20 cells in each group. Both targets enter each cell mean. Run run-20261020T1555Z; rows from `runs/*.jsonl`, plotted by `plot.py`.

P4: another prime

A0 and A11, m = 2 (t = 2–8) and m = 3 (t = 2–4): 40 systems, all completed. **Rule: prime-independent.** At all 20 shared cells the cell D equals the p_1 D, and the m = 3 gap holds (A11 at least 1 below A0 at t = 2, 3, 4). The widest matrix equals p_1's at every one of the 20 cells, to the column.

Z1: structured targets, $x_R = 0$

Each target brings its own curve, built around its points (Amendment 9). All 40 curves differ, and every target has $x_R = 0$, checked by the group law.

The construction failed at m = 2, t = 2. The 4 systems there (A0 and A11, 2 targets each) aborted after 50 331 712 trials each, with no target found. At t = 2, V has 4 elements and 6 pairs, and for this V no pair has $x_1 + x_2$ a non-zero square, which the m = 2 construction needs. The pair does not depend on the random draw, so the search could not succeed. By the stop rule that ended

the m = 2 series (Amendment 1 item 5). Amendment 10 re-ran m = 2 from t = 3, and records t = 2 as **not constructible for this V**. The 4 aborted rows stay committed.

With that, Z1 has 44 systems: 40 completed (m = 2, t = 3–9: 28; m = 3, t = 2–4: 12) and the 4 aborted. **Rule: target-independent.** At all 20 shared cells the cell D equals the generic D, and the m = 3 gap holds. The matrix widths are **not** equal to the generic ones:

arm, m	cells	width against generic	range
A0, m = 2	7	6 smaller, 1 larger	−21.8% to +2.5%
A0, m = 3	3	1 equal, 1 larger, 1 smaller	−1.3% to +2.3%
A11, m = 2	7	2 equal, 2 smaller, 3 larger	−0.3% to +3.2%
A11, m = 3	3	2 smaller, 1 larger	−21.0% to +0.6%

Over the 20 cells: 3 equal, 11 smaller, 6 larger, within −21.8% and +3.2%. This differs from the curve families of K1 (Amendment 8), where widths barely moved: a target with $x_R = 0$ removes terms from the summation polynomial, and the matrices change with it, in both directions.

M5: m = 5

A0 at m = 5, t = 3, the smallest cell (t = 2 has 4 elements, fewer than the 5 summands): **both systems stop at the 300 s budget**, at partial D = 8 and a widest matrix of 117 005 columns so far (315 s and 313 s). The partial D is already 1 above A0′s m = 4, t = 3 value (7). The cell does not complete, so there is no D and no σ at m = 5. A11 was not run (its form is written for m ≤ 4).

Predictions scorecard (Amendment 9)

prediction	observed	held?
P4: prime-independent	20 of 20 cells equal	yes
P4: the gap holds	3 of 3	yes
P4: `max_cols` equals p_1′s at every shared cell	20 of 20	yes
Z1: target-independent	20 of 20 cells equal	yes
Z1: the gap holds	3 of 3	yes
Z1 (Amendment 10): every t ≥ 3 cell constructs and completes	all 28 m = 2 systems and all 12 m = 3 systems	yes
M5: the cell does not complete in 300 s and 13 GB	both systems stop at 300 s	yes

Reading

- **Instances now compared.** D agrees with the first run's generic p_1 cells across a fourth prime of another 2-adic shape, two further curve families, and structured targets, as well as the earlier second prime, the tower engine and the narrow path. In each case where a cell was tested, D was equal and the A11-below-A0 gap held. This is the same pattern as in every earlier comparison.
- **Widths are less rigid than D.** They were equal across primes (P4, as for p_2) and across a curve family with the generic support (`a = −3`), nearly equal for `a = 0`, and changed by up to 22% for structured targets, where the support itself changes. So "cost is set by the system's shape" is, in these rows, a statement about D first, and about width only when the support is the same.
- **m = 5** is out of reach for the dense engine here: the smallest cell was already past 100 000 columns at its budget, with D one above m = 4′s.

What this still does not show

- **Not measured:** Edwards and binary-field curves; other structured targets (an x_R other than 0, or R of small order); primes with 2-adic valuation below 8; m ≥ 5 beyond its smallest cell; A11 at m = 5.
- **The structured target** is one kind, and its curves are built per target, so a structured-target cell differs from its generic comparison in both the target and the curve. The result says D is unchanged by this combination, not which part would matter alone.
- **The t = 2, m = 2 cell** was not constructible for the V this seed draws. A different seed could construct it.
- Two planted targets per cell, one cell per (arm, m, t), wall-clock on one container.

Deviations

The Z1 m = 2 series needed Amendment 10 (registered before its rows) after the t = 2 construction failure. No other deviation from Amendment 9.